Ewen LE CALLET

AI Research Intern – Efficient AI

Generative AI • Multimodal Models • Computational Efficiency • Research

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PROFILE

Final-year engineering student at Polytech Nantes, France, specialising in Multimedia Systems and Real-Time Computing. Seeking a **5–6-month AI research internship at Canva Beijing from March 2027**.

Research experience at Shanghai Jiao Tong University, with **two papers accepted at VCIP 2026**, including one as first author. Background in generative image quality assessment, multimodal AI and human visual perception, complemented by hands-on work on video token reduction, parameter-efficient fine-tuning and computationally constrained systems. Interested in **efficient generative AI for creative applications**, combining AI research with hands-on experience in audiovisual creation, video editing and image design.

RESEARCH EXPERIENCE

AI Research Intern – Shanghai Jiao Tong University (SJTU) *Shanghai, China, May–September 2026*
Multimedia Lab – Institute of Image Communication and Network Engineering

- **Generative image quality assessment:** investigated saliency bias in image editing quality assessment (IEQA) metrics, analysing discrepancies between predicted quality maps and human visual attention.
- **Human perception and statistical analysis:** analysed eye-tracking data to study how image editing changes visual attention, using statistical tests, saliency map comparisons and temporal analysis.
- **AI evaluator for audiovisual generation pipelines:** developing an AI-based evaluator specialised in AI-generated audiovisual production pipelines, covering scriptwriting, storyboarding, image generation and video generation. Ongoing first-author research, with a paper submission planned for late 2026.
- Conducted research across literature review, hypothesis development, experimental design, data analysis and scientific writing.

Research Publications

- **First author:** "Saliency Bias in Image Editing Quality Assessment" – accepted at the IEEE International Conference on Visual Communications and Image Processing **VCIP 2026**.
- **Co-author:** "The Influence of Image Editing on Visual Attention" – accepted at the IEEE International Conference on Visual Communications and Image Processing **VCIP 2026**.

SELECTED TECHNICAL PROJECTS

Multimodal AI – Video Token Minimisation and LoRA *2025*
Student Research Project – LS2N Laboratory
GitHub Repository

- **Video token reduction:** developed a tool to reduce the visual token cost of video inputs to multimodal models through temporal and semantic frame selection (CLIP, K-Means), scene change detection and spatial cropping.
- **Parameter-efficient fine-tuning:** explored LoRA (Low-Rank Adaptation) for adapting multimodal models to visual reasoning tasks.

Event-Based Vision – Efficient Embedded Detection *2026–2027*
Student Research Project – LS2N Laboratory

- Developing computationally efficient barcode detection and decoding algorithms operating directly on event-camera streams, targeting resource-constrained drone systems.
- Generating synthetic and real-world datasets using Blender simulations and robotic arm-based data acquisition.

Edge AI – Autonomous Wildlife Monitoring *2025–2026*
Student Project Lead – International team of 6

- Led the development of an autonomous wildlife monitoring prototype integrating AI-based species classification, multimodal sensing and automated reporting.

TECHNICAL SKILLS

AI & Deep Learning: PyTorch, TensorFlow, classical machine learning, CNNs, autoencoders, multimodal vision-language models, generative AI, LoRA, model evaluation and benchmarking.

AI Efficiency & Computing: Video token reduction, parameter-efficient fine-tuning, parallel programming, CUDA, embedded GPU platforms (NVIDIA Jetson), computational efficiency.

Computer Vision & Multimedia: Image and image editing quality assessment (IQA / IEQA), visual saliency, image detection and classification, event-based vision, image and video compression, synthetic dataset generation (Blender).

Programming & Systems: Python, C, Java, Assembly, RISC-V, Linux, Git.

Data Analysis & Research: NumPy, MATLAB, statistical analysis, experimental design, scientific literature review and writing.

PROFESSIONAL EXPERIENCE

R&D Intern – LCJ Capteurs *Vertou, France, June–August 2025*

Ultrasonic wind sensors

- Designed embedded signal processing algorithms for implementation on a dsPIC33 microcontroller.
- Extracted phase and amplitude from ultrasonic signals using the Hilbert transform under computational and memory constraints.

Observational Placements: La Pompadour Studio, motion design and post-production (Nantes, France, 2023); Environmental and Food Laboratory of Vendée (LEAV), France (2018).

EDUCATION

Polytech Nantes – Nantes Université *2024–2027*

Engineering Degree in Electronics and Digital Technologies, specialisation in Multimedia Systems and Real-Time Computing. Expected graduation: 2027.

Université d'Angers *2022–2024*

Integrated Preparatory Programme for Engineering Studies; mathematics, physics and engineering sciences.

Lycée Aimé Césaire, Clisson, France *2019–2022*

French Baccalauréat, Mathematics and Physics-Chemistry; awarded honours.

LANGUAGES & ADDITIONAL ACTIVITIES

Languages: French (native), English (TOEIC 900), Mandarin Chinese (beginner, currently learning).

Audiovisual Creation: Video editing, audiovisual production, image retouching and graphic design.

Other Activities: Basketball player and officially licensed referee (Loire-Atlantique and University Basketball League); drums and guitar. UNICEF volunteer in communications and graphic design.